

## NEWCASTLE INTL, United Kingdom

WMO: 032433

Lat: 55.038N

Lon: 1.692W

Elev: 266

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>24.8 | (c)<br>28.0 | (d)<br>20.8                   | (e)<br>15.8 | (f)<br>28.3 | (g)<br>23.0 | (h)<br>17.5 | (i)<br>30.4 | (j)<br>37.5           | (k)<br>46.9 | (l)<br>32.5 | (m)<br>46.0 | (n)<br>3.0            | (o)<br>280 | (p)<br>0.619 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.5                   | 73.6            | 63.3 | 71.2 | 62.4 | 68.2 | 60.5 | 65.1                | 71.7 | 63.3 | 69.2 | 61.6 | 66.7 | 8.1                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.6                            | 85.8 | 68.1 | 60.9 | 80.6 | 66.0 | 59.2 | 75.8 | 64.2 | 30.1          | 71.4 | 28.8 | 68.7 | 27.6 | 66.9 | 75.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 27.3          | (b) 23.5 | (c) 20.5 | (e) 19.1                   | (f) 80.1 | (g) 4.2            | (h) 2.6 | (i) 16.1                                           | (j) 82.0 | (k) 13.6   | (l) 83.5 | (m) 11.3   | (n) 85.0 | (o) 8.2    | (p) 86.9 |    |    |
| (a) 27.3          | (b) 23.5 | (c) 20.5 | (e) 19.0                   | (f) 69.0 | (g) 4.3            | (h) 2.2 | (i) 15.9                                           | (j) 70.6 | (k) 13.5   | (l) 71.9 | (m) 11.1   | (n) 73.1 | (o) 8.0    | (p) 74.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 48.3   | 39.2  | 40.0  | 42.1  | 45.6  | 50.5  | 55.4  | 59.2  | 59.1  | 55.2  | 49.8  | 43.5  | 39.6  |  |  |
|                                                                        | DBStd       | 8.90   | 5.36  | 5.47  | 5.36  | 5.14  | 5.24  | 4.52  | 4.32  | 4.06  | 4.28  | 4.97  | 5.49  | 6.39  |  |  |
|                                                                        | HDD50       | 1676   | 336   | 283   | 248   | 149   | 57    | 6     | 0     | 0     | 6     | 64    | 201   | 326   |  |  |
|                                                                        | HDD65       | 6102   | 800   | 701   | 709   | 582   | 449   | 289   | 186   | 189   | 294   | 471   | 644   | 788   |  |  |
|                                                                        | CDD50       | 1065   | 0     | 2     | 5     | 17    | 73    | 169   | 286   | 282   | 163   | 58    | 8     | 2     |  |  |
|                                                                        | CDD65       | 16     | 0     | 0     | 0     | 0     | 0     | 2     | 7     | 6     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 84     | 0     | 0     | 0     | 0     | 1     | 11    | 39    | 29    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 5      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 3     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 8.3    | 10.0  | 10.0  | 9.7   | 8.3   | 7.9   | 7.3   | 6.7   | 7.0   | 7.4   | 7.8   | 8.4   | 9.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 27.5   | 2.2   | 1.8   | 1.7   | 2.0   | 2.0   | 2.4   | 2.5   | 2.6   | 2.2   | 2.6   | 3.0   | 2.6   |  |  |
|                                                                        | PrecMax     | 40.8   | 5.7   | 3.9   | 3.8   | 5.7   | 4.2   | 6.7   | 6.0   | 6.4   | 5.6   | 5.4   | 6.5   | 5.7   |  |  |
|                                                                        | PrecMin     | 20.6   | 0.7   | 0.3   | 0.4   | 0.3   | 0.5   | 0.6   | 0.4   | 0.5   | 0.7   | 0.6   | 0.9   | 1.1   |  |  |
|                                                                        | PrecStd     | 4.5    | 1.1   | 1.0   | 1.0   | 1.6   | 1.0   | 1.6   | 1.3   | 1.2   | 1.3   | 1.1   | 1.4   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 55.0  | 56.8  | 59.1  | 65.9  | 71.8  | 75.3  | 78.6  | 78.4  | 73.5  | 64.8  | 59.0  | 55.8  |  |  |
|                                                                        |             | MCWB   | 50.9  | 51.6  | 51.3  | 56.3  | 61.7  | 63.7  | 65.1  | 66.0  | 62.9  | 58.8  | 55.5  | 52.4  |  |  |
|                                                                        | 2%          | DB     | 51.9  | 53.3  | 55.4  | 60.6  | 66.4  | 70.1  | 73.7  | 72.0  | 68.3  | 62.2  | 55.7  | 53.6  |  |  |
|                                                                        |             | MCWB   | 48.4  | 49.3  | 50.3  | 53.1  | 57.3  | 60.8  | 63.4  | 63.4  | 60.7  | 57.7  | 52.5  | 50.4  |  |  |
|                                                                        | 5%          | DB     | 49.8  | 50.4  | 53.4  | 57.2  | 62.7  | 67.6  | 71.2  | 69.6  | 64.8  | 59.4  | 53.7  | 51.7  |  |  |
|                                                                        |             | MCWB   | 46.7  | 46.8  | 48.6  | 50.8  | 55.1  | 59.4  | 62.6  | 62.3  | 58.3  | 55.7  | 50.7  | 48.6  |  |  |
|                                                                        | 10%         | DB     | 47.8  | 48.4  | 50.4  | 55.0  | 59.4  | 64.4  | 68.0  | 66.5  | 62.8  | 57.5  | 51.8  | 48.6  |  |  |
|                                                                        |             | MCWB   | 45.4  | 45.3  | 46.3  | 49.4  | 52.9  | 57.2  | 60.7  | 59.6  | 57.3  | 53.9  | 49.0  | 46.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 51.3  | 52.1  | 53.6  | 56.8  | 62.4  | 66.2  | 67.3  | 67.6  | 64.9  | 60.1  | 55.9  | 53.8  |  |  |
|                                                                        |             | MCDB   | 54.4  | 55.6  | 58.7  | 63.9  | 71.0  | 74.1  | 74.8  | 72.9  | 71.6  | 63.6  | 58.3  | 56.1  |  |  |
|                                                                        | 2%          | WB     | 49.1  | 49.7  | 51.1  | 53.8  | 58.9  | 62.2  | 65.1  | 64.9  | 62.0  | 58.3  | 53.5  | 51.1  |  |  |
|                                                                        |             | MCDB   | 51.7  | 52.9  | 54.6  | 59.3  | 65.3  | 68.2  | 71.7  | 71.1  | 67.1  | 61.4  | 55.9  | 53.5  |  |  |
|                                                                        | 5%          | WB     | 46.9  | 47.7  | 49.1  | 51.6  | 56.1  | 59.8  | 63.2  | 63.0  | 59.8  | 56.5  | 51.6  | 49.1  |  |  |
|                                                                        |             | MCDB   | 49.2  | 50.9  | 52.5  | 56.4  | 61.5  | 65.7  | 69.1  | 68.3  | 63.9  | 59.5  | 53.7  | 51.4  |  |  |
|                                                                        | 10%         | WB     | 44.9  | 45.5  | 47.0  | 49.8  | 53.9  | 57.9  | 61.4  | 61.2  | 58.2  | 54.7  | 49.7  | 46.7  |  |  |
|                                                                        |             | MCDB   | 46.8  | 48.0  | 50.5  | 54.0  | 59.1  | 63.5  | 66.8  | 65.4  | 62.0  | 57.6  | 51.3  | 48.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.3   | 9.9   | 11.8  | 13.4  | 14.1  | 13.9  | 14.5  | 14.3  | 13.3  | 11.5  | 9.7   | 8.9   |  |  |
|                                                                        |             | MCDBR  | 10.8  | 11.7  | 15.0  | 17.7  | 19.4  | 19.5  | 19.3  | 17.9  | 17.2  | 13.5  | 10.7  | 11.0  |  |  |
|                                                                        |             | MCWBR  | 9.8   | 9.7   | 10.5  | 11.6  | 12.2  | 11.9  | 10.9  | 10.6  | 10.8  | 10.4  | 9.5   | 10.0  |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.4  | 11.3  | 13.5  | 16.5  | 18.3  | 17.7  | 17.3  | 15.9  | 15.2  | 12.7  | 10.4  | 10.9  |  |  |
|                                                                        |             | MCWBR  | 9.9   | 9.9   | 10.6  | 11.6  | 12.3  | 11.8  | 10.8  | 10.4  | 10.7  | 10.5  | 9.7   | 10.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.316 | 0.333 | 0.364 | 0.387 | 0.385 | 0.387 | 0.388 | 0.378 | 0.358 | 0.342 | 0.316 | 0.299 |  |  |
|                                                                        | taud        |        | 2.310 | 2.380 | 2.310 | 2.270 | 2.324 | 2.346 | 2.349 | 2.391 | 2.449 | 2.451 | 2.369 | 2.277 |  |  |
|                                                                        | Ebn at Noon |        | 194   | 237   | 254   | 264   | 272   | 272   | 270   | 265   | 256   | 231   | 193   | 171   |  |  |
|                                                                        | Edh at Noon |        | 18    | 23    | 31    | 37    | 37    | 37    | 36    | 33    | 27    | 22    | 16    | 15    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 178   | 380   | 740   | 1203  | 1560  | 1559  | 1494  | 1237  | 880   | 478   | 226   | 132   |  |  |
|                                                                        | RadStd      |        | 23    | 41    | 89    | 117   | 89    | 129   | 124   | 83    | 75    | 42    | 20    | 14    |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (47 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page